Experiment - 05.

- Aim: Perform the data classification using classification algorithm or perform the data clustering using clustering algorithm.

- Theory :

(i) Introduction -

* Business Intelligence refers to technologies, processes and tools used to transform raw data into meaningful information for strategic decision making.

- Two important data mining techniques in BI systems are:
  - Data classification.
  - Data clustering.

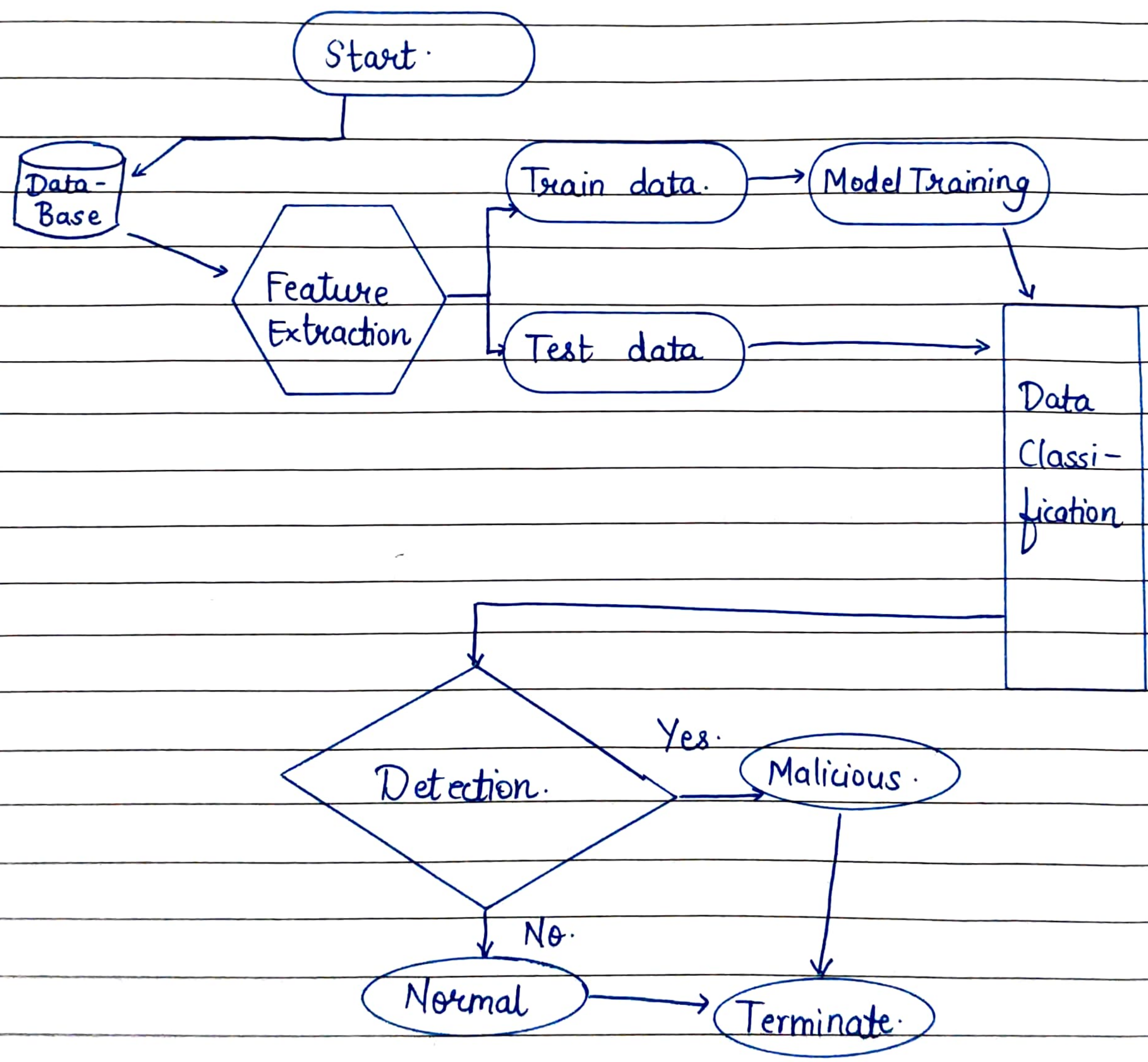
- These techniques help organizations identify patterns, predict trends and support manegerial decisions.

- In BI, classification is used to -
  - Assign business entities into predefined categories.
  - Support predictive decision making.
  - Automate business rule enforcement.

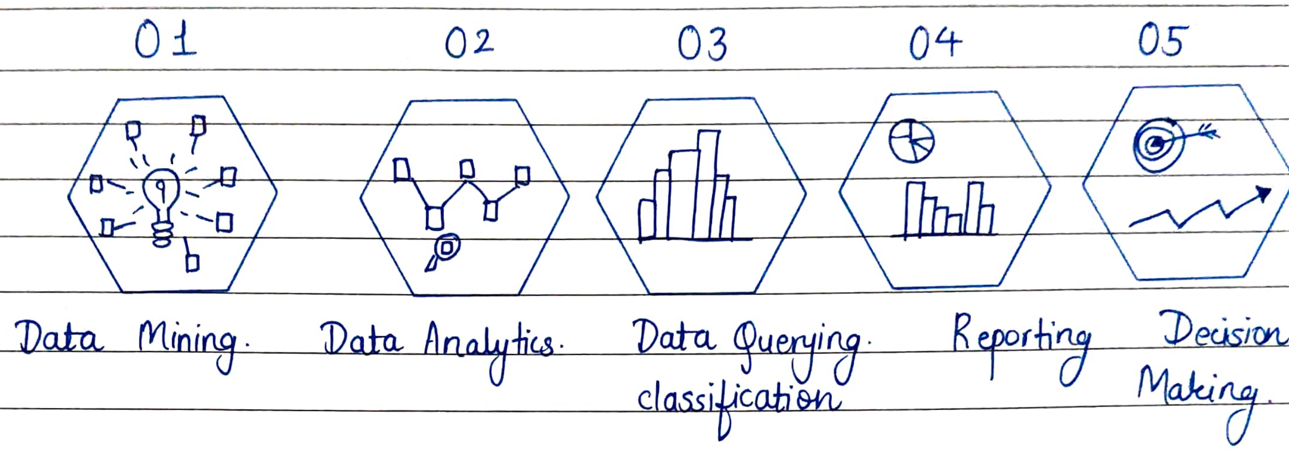
- Classification is viewed as decision support tool.

(2). Business Oriented Examples –

- Classifying customers :
    - High value.
    - Medium value.
    - Low - value.
- Approving or rejecting loan applications.
- Classifying transactions as:
    - Fraudulent
    - Genuine.
- Categorizing products based on sales performance.



BI Process.



- **Flow explanation** -

1. Data collection - Data extracted from ERP, CRM, HR systems.
2. Data warehousing - Integrated and stored in data warehouse.
3. Data cleaning and transformation -
   - remove inconsistencies.
   - standardize business rules.
4. Apply classification rules / Models. -
   - Based on historical business models.
5. Generate reports and dashboards -
   - show category distribution.
6. Managerial decision making -
   - Strategic planning.
   - Policy formation.

(3). Role of classification in decision support systems (DSS) -

Classification helps -
- Risk assessment.
- Credit scoring.
- Market prediction.
- Sales forecasting categories.

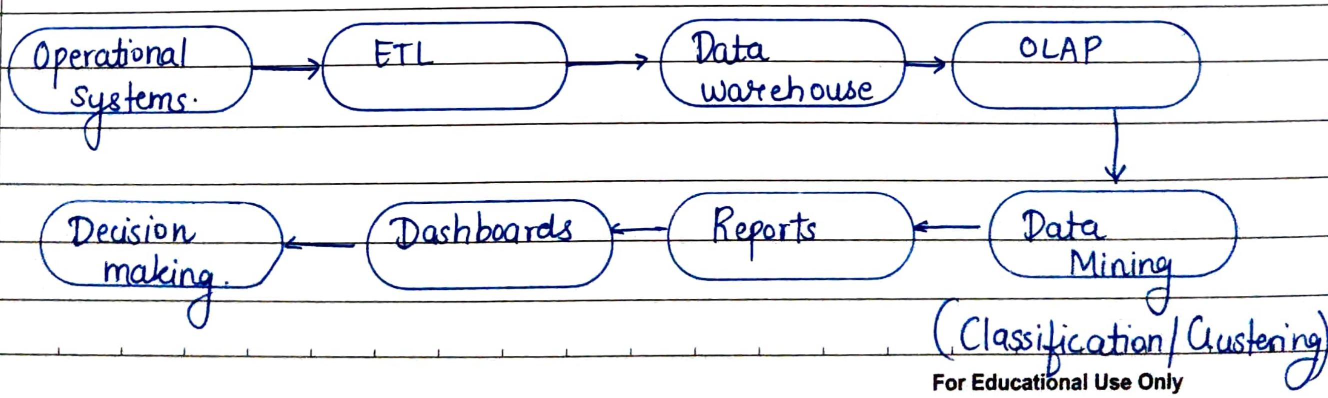
It improves -
- Accuracy of business predictions.
- Speed of decision making.
- Reduction of Human Bias.

(4) Strategic importance of clustering -

- Clustering supports -
  - Targetted marketting campaigns.
  - Personalized recommendations.
  - Inventory planning.
  - Regional expansion strategy.

- It helps management answer questions like -
  - Which customers are most profitable.
  - Which regions perform similarly.
  - Which products are purchased together.

(5). Flow -



(6). Classification Algorithm – Decision Tree –

A decision tree splits data based on feature values to classify instances. It selects features using information gain or gini index.

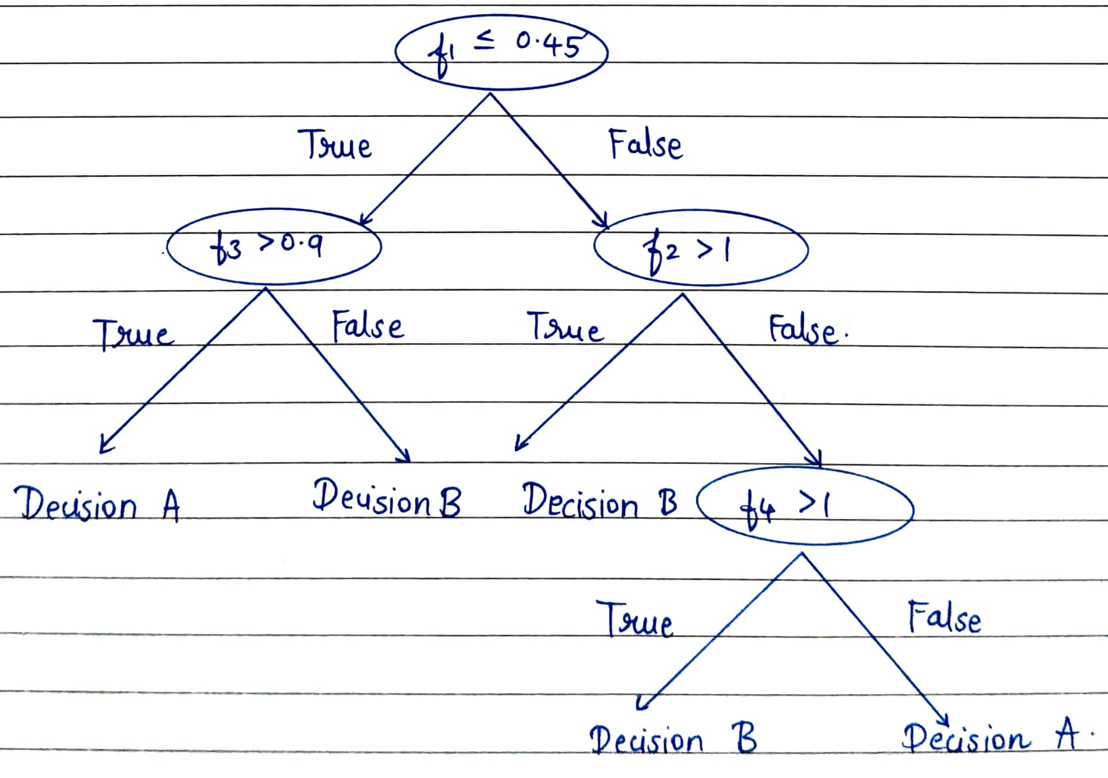
Entropy formula – $$\text{Entropy}(S) = -\sum_{i=1}^{c} p_i \log_2 p_i$$

Where,
- p_i = probability of class i
- c = number of classes.

Information gain –

$$IG(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

* Decision tree structure –



(7) **Clustering Algorithm - K-means clustering**

K-means partitions data into k clusters by minimizing intra-cluster variance.

K-means objective function

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

Where,
- μ_i = centroid of cluster i
- $||x - \mu_i||^2$ = Euclidean distance.

K means Algorithm steps -
- Choose K centroids randomly.
- Assign each point to nearest centroid.
- Recompute centroids.
- Repeat until convergence.

(8) **Comparison of Classification and clustering.**

	Feature.	Classification.	Clustering.
1.	Learning type.	- Supervised	Unsupervised
2.	Data	- Labeled.	Unlabeled.
3.	Output.	- Class labels	Cluster groups.
4.	Example Algorithm.	- Decision Tree.	K-means.

(9) **CONCLUSION:** Thus we successfully segregated data through classification algorithm and clustering for business purposes.

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